

Lecture 2

Describing univariate distributions

Sean P. Corcoran

LPO.8800: Statistical Methods for Education Research



Introduction



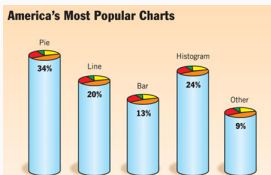
- Descriptive vs. inferential statistics
- Basic concepts: outcomes, variables, unit of observation, population, sample
- Measurement scales
 - quantitative or categorical
 - nominal, ordinal, interval, ratio
 - discrete vs. continuous



1

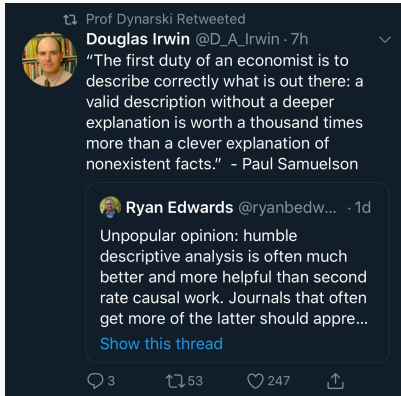
This lecture

- Describing univariate distributions: categorical and quantitative
- Measures of central tendency, variability, skewness, relative position
- Data transformations



2

Importance of descriptive analyses



3

Importance of descriptive analyses

See the IES report by Loeb et al. (2017), which describes the role of descriptive analysis in education and social science.

Key Themes

- Descriptive analysis characterizes the world or a phenomenon—answering questions about who, what, where, when, and to what extent. Whether the goal is to identify and describe trends and variation in populations, create new measures of key phenomena, or describe samples in studies aimed at identifying causal effects, description plays a critical role in the scientific process in general and education research in particular.
- Descriptive analysis stands on its own as a research product, such as when it identifies socially important phenomena that have not previously been recognized. In many instances, description can also point toward causal understanding and to the mechanisms behind causal relationships.
- No matter how significant a researcher's findings might be, they contribute to knowledge and practice only when others read and understand the conclusions. Part of the researcher's job and expertise is to use appropriate analytical, communication, and data visualization methods to translate raw data into reported findings in a format that is useful for each intended audience.



4

Some examples of excellent, influential descriptive studies in education:

- Arnold et al. (2009) - on “summer melt”
- Scott-Clayton (2012) - changes over time in undergraduates’ propensity to work while in school
- Reardon (2011) - on changes in the academic achievement gap between high- and low-income students
- Lankford, Loeb, & Wyckoff (2002) - on the distribution of teacher qualities across New York State districts and schools
- Hoxby & Avery (2003) - on the “missing one-offs”
- Murnane (2013) - on long-run trends in U.S. HS graduation rates



Frequency distributions



- A **frequency distribution** is a table showing the number (count) of occurrences of each unique outcome in the data.
- The **relative frequency** of an outcome or category is the *proportion or percentage* of all observations in that category. (Must sum to one, or 100%).



6

Frequency distributions

Example: test scores for 16 students

Table 1: Classroom test scores

Score	Frequency (Count of students)
5	2
10	3
12	1
15	4
20	4
25	2
Total	16



7

Frequency distributions

Example: test scores for 16 students

Table 2: Classroom test scores

Score	Frequency (Count of students)	Relative Frequency (Percentage)
5	2	$2/16 * 100 = 12.5\%$
10	3	$3/16 * 100 = 18.75\%$
12	1	$1/16 * 100 = 6.25\%$
15	4	$4/16 * 100 = 25.0\%$
20	4	$4/16 * 100 = 25.0\%$
25	2	$2/16 * 100 = 12.5\%$
Total	16	$16/16 * 100 = 100\%$



8

Frequency distributions

Side note on terminology:

- The fractions in the above relative frequency distribution: $2/16$, $3/16$, $1/16$ —or, 0.125, 0.1875, 0.0625, etc.—are **proportions**: the frequency of cases in a given category divided by the total number of cases in all categories. Ranges between zero and one.
- Multiply by 100 to get **percentages** (12.5%, 18.75%, 6.25%, etc.)
- Relative frequencies can be expressed either way.

Pro tip: see handout on **rounding conventions**. Percentages are typically rounded to one decimal place; proportions typically rounded to three.



9

- Because frequency distributions list every distinct value in the data, they are only practical for variables with a limited number of unique values
 - Categorical variables
 - Discrete quantitative variables with a manageable number of values
- It is possible to *group* variables with many distinct outcomes into smaller categories (shown later)
 - Continuous quantitative variables



10

Frequency distributions

Easy to generate in Stata using `tabulate`

```
. tabulate region
```

geographic region of school	Freq.	Percent	Cum.
northeast	106	21.20	21.20
north central	151	30.20	51.40
south	150	30.00	81.40
west	93	18.60	100.00
Total	500	100.00	

Note the *cumulative* column is not very meaningful here (a categorical, non-ordered variable).



11

Example:

```
summarize varlist [if] [in] [weight] [,options]
```

- `summarize`: the command
- *varlist*: terms in italics are things you provide
- Syntax in brackets [] is *optional*
- [,*options*]: Most commands have other options that are specified *after* a comma, at the very end
- [*if*]: execute the command using only observations where a condition is true
- [*in*]: execute the command for a certain subset of observation numbers



12

Frequency distributions

To show the count of missing values in Stata, include the `missing` option: `tabulate varname, missing`

```
. tabulate parmar18, missing
```

parents' marital status in eighth grade	Freq.	Percent	Cum.
divorced	26	5.20	5.20
widowed	6	1.20	6.40
separated	8	1.60	8.00
never married	5	1.00	9.00
marriage-like relationship	5	1.00	10.00
married	427	85.40	95.40
.	23	4.60	100.00
Total	500	100.00	



13

Also see the user-created **fre** command which shows the relative frequency with and without missing values:

```
. fre parmar18
```

parmar18 — parents' marital status in eighth grade

		Freq.	Percent	Valid	Cum.
Valid	1 divorced	26	5.20	5.45	5.45
	2 widowed	6	1.20	1.26	6.71
	3 separated	8	1.60	1.68	8.39
	4 never married	5	1.00	1.05	9.43
	5 marriage-like relationship	5	1.00	1.05	10.48
	6 married	427	85.40	89.52	100.00
	Total	477	95.40	100.00	
Missing	.	23	4.60		
Total		500	100.00		

fre is also nice in that it shows you both variable *values* and *labels*.



14

Frequency distributions

The **table** command provides a simpler frequency distribution (without the percent or cumulative percent).

```
. table parmar18
```

parents' marital status in eighth grade	Freq.
divorced	26
widowed	6
separated	8
never married	5
marriage-like relationship	5
married	427

The **table** command was substantially overhauled in Stata 17, with lots of options for customizing format and cell contents.



15

Note the categories used in each of these variables are mutually exclusive and collectively exhaustive:

- **mutually exclusive:** being in one category precludes being in another
- **collectively exhaustive:** all possible categories are represented by the defined categories



16

Grouped frequency distributions

- Again, frequency distributions are less useful for variables with *many* distinct possible values (e.g. continuous variables)
- For continuous variables, one could create a smaller number of equal width groups (bins, or intervals) and then create a frequency distribution for this grouped (“re-coded”) variable. (This is what histogram does behind the scenes).



17

- NELS example: *unitmath* is the number of units of high school math taken, and ranges from 1-6. Includes many fractional units.
- Can set up groups or intervals, for example:
 - $1 \leq x < 2$
 - $2 \leq x < 3$
 - $3 \leq x < 4$, and so on
- Later: how to re-code variables in this way (a kind of data transformation)



Bar graphs



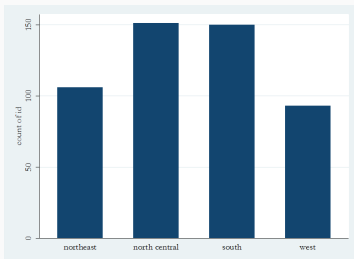
- **Bar graphs** are a visual way to display frequency or relative frequency distributions
- In Stata: `graph bar` can be used for frequency (or relative frequency) distributions. An alternative is `histogram` with the `discrete` option.
 - `(count)` for frequency
 - `(percent)` for relative frequency
- Try: `region`, `parmarl8`, `advmath8`



19

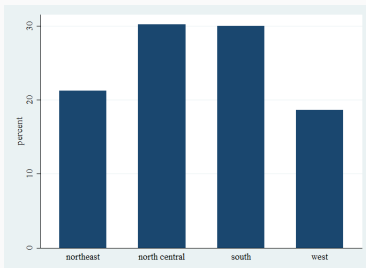
Bar graphs

`graph bar (count), over(region)`



20

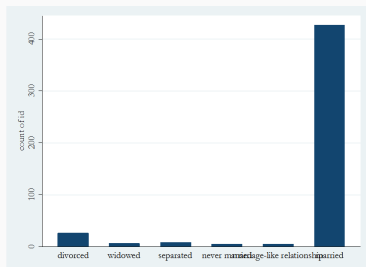
`graph bar (percent), over(region)`



21



`graph bar (count), over(parmar18)`

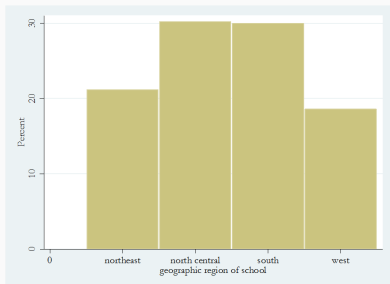


22



Bar graphs

```
histogram region, discrete percent gap(2) xlabel( ,  
valuelabel)
```



23



Bar graphs

- Word of caution with bar graphs: always check where vertical axis begins—does it begin at zero? Avoid misleading scales
- Note “bar graphs” (as opposed to histograms) have gaps between the bars, suggestive of distinct categories
- Note The `gap(2)` option in `histogram` forces a gap of size 2

24



Pie graphs



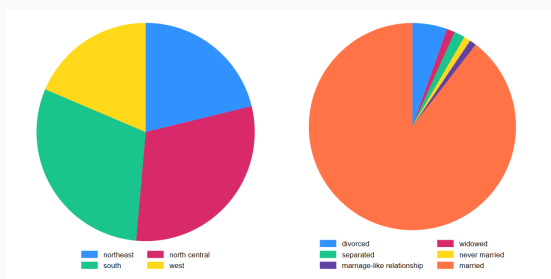
Pie graphs

Pie graphs can be used to show the relative frequency of a variable

- These only make sense when the variable has *collectively exhaustive* (and a limited number of) categories
- In Stata: `graph pie`
- Try: `region, parmarl8`



```
graph pie, over(region)
```

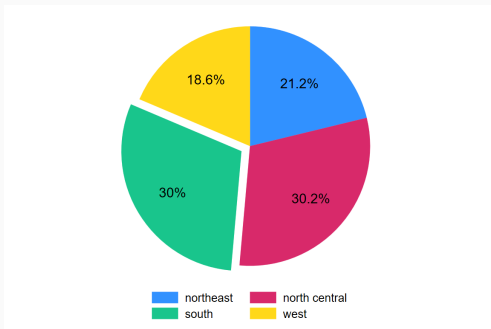


26



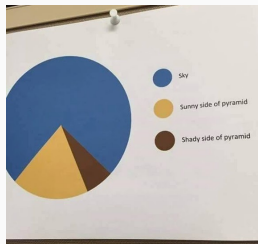
Pie graphs (options)

```
graph pie, over(region) plabel(_all percent,  
size(medlarge)) pie(3,explode)
```



27





Histograms



A **histogram** is a bar graph where the height of each bar represents the count (or percent) of observations within a given *range* of values called an **interval** or **bin**

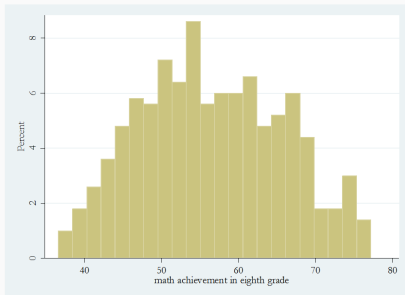
- Number of bins determined by default in Stata, but can be adjusted
- Obviously makes sense only for interval (or ratio) measured variables, where a range of values is meaningful
- In Stata: `histogram`



29

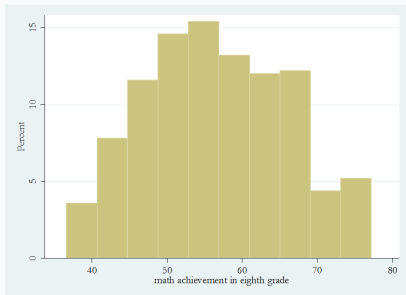
Histograms

`histogram achmat08, percent`



30

```
histogram achmat08, percent bin(10)
```



Stem-and-leaf plot



A **stem-and-leaf** plot is similar to a histogram, but provides a bit more detail on specific values in the data

- Leading digits are called “stems”
- Trailing digits are called “leaves”
- The number of leaves corresponds to the frequency of a particular value

32



Stem-and-leaf plot

Stem-and-leaf plot for achmat08 (math achievement in eighth grade)

achmat08 rounded to nearest multiple of .1
plot in units of .1

36*	6
37*	122
38*	478
39*	0357
40*	0225555799
41*	34689
42*	0335577
43*	000133445689
44*	00012244467788
45*	234467778899
46*	3445555667788889
47*	0022344456677888
48*	00000133667889
49*	022233444666889
50*	001112356677888899
51*	0000111233334455779
52*	111112223334455666889
53*	000022456677889
54*	000011223334445555778888999
55*	00000003345678899
56*	11223346777888889
57*	0000112233344444778
58*	001113334677888899999
59*	034446679
60*	001223334556777888
61*	000112333345689
62*	000111123334555889
63*	2223677777779
64*	0112566788899
65*	223455677889
66*	0001122333344667799
67*	122333456677889
68*	0022346666899
69*	0122446779
70*	0114899
71*	389
72*	146
73*	3335788
74*	02233334
75*	00014
76*	
77*	222222

33



- In the above example, Stata rounded to the nearest 0.1
- First two digits used as stem, and final digit used as leaf
- Leaves correspond to frequency of particular values
- Compare the shape of the stem-and-leaf plot for *actmat08* to that of the histogram



Shape of distributions

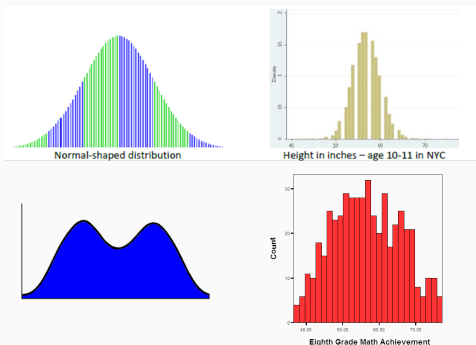


- The histogram (and stem-and-leaf plots) are revealing about the *shape* of a distribution—i.e. which values tend to be more or less frequent
- A distribution is **symmetric** if the distribution of outcomes is identical (or approximately identical) on either side of its central value
- Examples: normal distribution (bell curve), U-shaped distribution, bi-modal distribution, uniform distribution



35

Shape of distributions



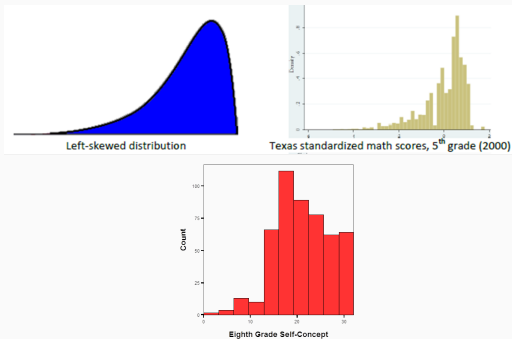
36

- A distribution is **skewed left** (or **negatively skewed**) if the distribution has a long tail to the left of its central value
- A distribution is **skewed right** (or **positively skewed**) if the distribution has a long tail to the right of its central value



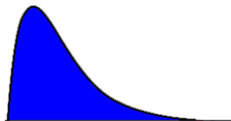
37

Left-skewed distributions



38

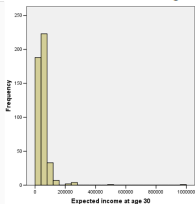




Right-skewed distribution



Weight in pounds – age 10-11 in NYC



Measures of central tendency



Measures of central tendency characterize the “center” or “location” of a distribution, its “typical” or “expected” value. Examples:

- Mean
- Median
- Mode

40



Mean

The **mean** (or *average*) adds all of the observed values and divides by the number of observations n

- Let $x_1, x_2, x_3, \dots, x_n$ represent the n values of a variable x (x_i is the i th observation, and i is the *index*)
- Then the **mean** is: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

41



Example: calculating the mean PISA math score for 15 highest income countries (2022)

Table 3: PISA math scores, 2022

	GNI per cap	Math (x)
Singapore	81,760	575
Switzerland	110,330	508
Canada	56,420	497
Netherlands	68,530	493
Ireland	87,360	492
Denmark	77,190	489
Belgium	59,500	489
Australia	64,120	487
Austria	60,360	487
Sweden	63,010	482
Germany	60,200	475
Norway	97,310	468
United States	88,810	465
Iceland	89,220	459
Qatar	74,330	414
Σx		7,280



42

Example: calculating the mean PISA math score for 15 highest income countries (2022)

$$\bar{x} = \frac{\sum x_i}{n} = \frac{7280}{15} = 485.3$$



43

The mean in Stata can be calculated using several commands, including `summarize` (or `sum`).

```
. sum achmat08
```

Variable	Obs	Mean	Std. Dev.	Min	Max
-----+-----					
achmat08	500	56.59102	9.339608	36.61	77.2

```
. sum expinc
```

Variable	obs	Mean	Std. Dev.	Min	Max
-----+-----					
expinc30	459	51574.73	58265.76	0	1000000

44



The mean of a categorical variable is usually meaningless, except in the case of a *dichotomous* variable coded 0-1, in which case the mean is the proportion equal to 1:

```
. sum advmath8
```

Variable	Obs	Mean	Std. Dev.	Min	Max
-----+-----					
advmath8	491	.4602851	.4989286	0	1

45



The mean is highly influenced by extreme values or **outliers**, observations that fall well above or well below the bulk of the data.

- Example 1 ($n = 15$)
 - 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5
 - mean = $(55/15) = 3.67$
- Example 2 ($n = 15$)
 - 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5000000
 - mean = $(5000050/15) = 333,336.67$



The mean can be characterized as the “center of gravity,” or balance point, of the distribution. It is the point at which the sum of the distances from the mean for observations *above* the mean equal the sum of the distances from the mean for observations *below* the mean.



Table 4: Deviations above and below the mean

	GNI per cap	Math (x)	$x - \bar{x}$	Totals
Singapore	81,760	575	89.7	
Switzerland	110,330	508	22.7	
Canada	56,420	497	11.7	
Netherlands	68,530	493	7.7	
Ireland	87,360	492	6.7	
Denmark	77,190	489	3.7	
Belgium	59,500	489	3.7	
Australia	64,120	487	1.7	
Austria	60,360	487	1.7	149.0
Sweden	63,010	482	-3.3	
Germany	60,200	475	-10.3	
Norway	97,310	468	-17.3	
United States	88,810	465	-20.3	
Iceland	89,220	459	-26.3	
Qatar	74,330	414	-71.3	-149.0

48



The least squares principle:

- The *average deviation* of x from its mean will always be zero. That is, the sum of negative deviations from the mean will always equal the sum of positive deviations from the mean.
- The mean is the point in a distribution around which the total variation is minimized (as indicated by the *squared* differences): $\sum (x_i - \bar{x})^2$ (proof later)



The **median** is the observation that falls in the middle of the data, when the observations are ordered from lowest to highest values.

- When n is *odd*: a single value will fall in the middle
- When n is *even*: the median is the midpoint of the two middle values
- Alternatively, index the ordered n values from 1 to n . The median will be the value with index $(n + 1)/2$



50

- Example 1 ($n = 15$):
 - 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5
 - median = 4
- Example 2 ($n = 6$):
 - 0, 1000, 1000, 5000, 6000, 8000
 - median = $(1000 + 5000)/2 = 3000$



51

The median in Stata can be found using several commands, including `summarize` (or `sum`) with the `detail` option:

```
. sum achmat08, detail
```

math achievement in eighth grade				
	Percentiles	Smallest		
1%	38.55	36.61		
5%	41.89	37.14		
10%	44.185	37.2	obs	500
25%	49.42	37.24	Sum of wgt.	500
50%	56.18		Mean	56.59102
75%	63.74	Largest	Std. Dev.	9.339608
90%	68.935	77.2	Variance	87.22827
95%	73.33	77.2	Skewness	.1133238
99%	77.2	77.2	Kurtosis	2.242742

50th percentile = median

52



Because the median is simply the middle value, it is insensitive (“robust”) to extreme values or outliers in the distribution

- Example 1 ($n = 15$):
 - 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5000000
 - median = 4

53



The **mode** is the outcome that occurs most often in a distribution

- Most appropriate for highly discrete variables, such as categorical variables
- Variables with lots of unique values (such as continuous variables) tend to have few repeats, and thus the mode is not that meaningful
- There is no command in Stata specifically for obtaining the mode. However, can use a frequency distribution or other combinations of commands (like `egen`). See also the user-written command `modes` (type `search modes`)



54

Mode

```
. ** Find mode using tabulate with sort (for descending sort)
```

```
. tabulate famsize, sort
```

family size	Freq.	Percent	Cum.
4	199	39.80	39.80
5	142	28.40	68.20
6	55	11.00	79.20
3	52	10.40	89.60
7	21	4.20	93.80
9	13	2.60	96.40
2	9	1.80	98.20
8	9	1.80	100.00
Total	500	100.00	

```
. ** Find mode using egen (note you will get a message if >1 mode)
```

```
. egen mode=mode(famsize)
```

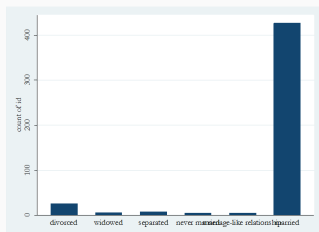
```
. table mode
```

mode	Freq.
4	500



55

“Married” is the modal parents’ marital status. One might say the “typical” 8th grade student in the NELS has married parents.



56



North Central is (technically) the modal region. But region here is more accurately described as *bimodal*—it has a distribution with two values that occur most often (North Central and South).

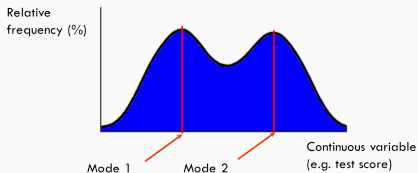
```
. tabulate region
```

geographic region of school	Freq.	Percent	Cum.
northeast	106	21.20	21.20
north central	151	30.20	51.40
south	150	30.00	81.40
west	93	18.60	100.00
Total	500	100.00	

57



Another *bimodal* distribution



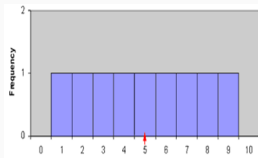
58



Mode

Some problems with using the mode:

- It is not very useful for “flat” distributions (e.g. the *uniform* distribution)
- Example 1: 1, 1, 2, 2, 3, 3, 4, 4, 5, 5
- Example 2:



59



How will the mode, median, and mean usually compare? It depends on the shape of the distribution.

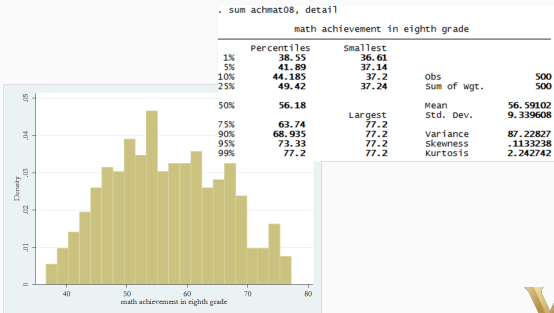
- For *symmetric* distributions the median $\approx$ mean
- If symmetric and *unimodal*, median $\approx$ mean $\approx$ mode (when the mode is meaningful, distributions with a limited number of unique values)



60

Comparing measures of central tendency

Example: note the mode of this distribution is 72 (not shown). Illustrates the problem of using the mode with a continuous variable.



61

Non-symmetric distributions:

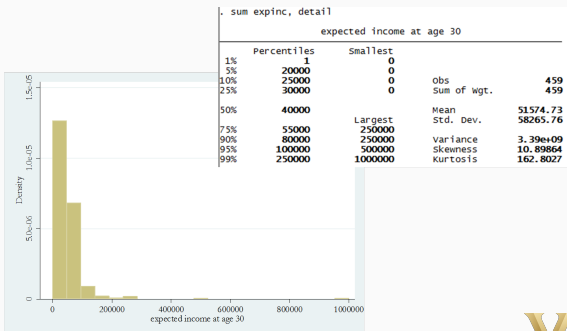
- For *right-skewed* distributions the mean $>$ median
- For *left-skewed* distributions the mean $<$ median



62

Comparing measures of central tendency

A right-skewed variable with some large positive outliers:

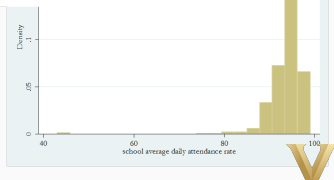


63

A left-skewed variable with some low-value outliers:

```
. sum schatrrt, detail
```

school average daily attendance rate				
	Percentiles	Smallest		
1%	80	43		
5%	88	45		
10%	90	75	obs	417
25%	93	78	Sum of wgt.	417
50%	95		Mean	93.64988
75%	96	Largest	Std. Dev.	4.690642
	98			
90%	97	99	Variance	22.00212
95%	98	99	Skewness	-6.212213
99%	98	99	Kurtosis	62.38636



64

Measuring the Black-white wealth gap

The True Cost of Closing the Racial Wealth Gap

<https://www.nytimes.com/2021/04/30/business/racial-wealth-gap.html>

Using data from the 2019 Survey of Consumer Finances:

- Median Black household wealth: \$24,100
- Median white household wealth: \$188,200
- Gap: **\$164,100**
- Mean Black household wealth: \$142,500
- Mean white household wealth: \$983,400
- Gap: **\$840,000**
- 97% of white households' total wealth is held by households above the median

65

An alternative command in Stata for measures of central tendency and other statistics:

```
tabstat achrdg*, stat(mean p50 n)
```

```
tabstat achrdg*, stat(mean p50 n) col(stat)
```

```
. tabstat achrdg*, stat(mean p50 n)
```

Stats	achrdg08	achrdg10	achrdg12
Mean	56.04906	56.11404	55.60188
p50	56.445	57.545	57.005
N	500	500	500

```
. tabstat achrdg*, stat(mean p50 n) col(stat)
```

Variable	Mean	p50	N
achrdg08	56.04906	56.445	500
achrdg10	56.11404	57.545	500
achrdg12	55.60188	57.005	500



On summations



The mean is written as: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

- $\sum$ is the summation operator
- i is the index of summation
- 1 and n are the lower and upper limit of the summation (i.e., summing the numbers x_i for all values of i from 1 to n)



67

Three properties of the summation operator

The summation operator has three properties:

- For any constant c : $\sum_{i=1}^n c = nc$
- For any constant c : $\sum_{i=1}^n cx_i = c \sum_{i=1}^n x_i$
- For any constants a and b :
 $\sum_{i=1}^n (ax_i + by_i) = a \sum_{i=1}^n x_i + b \sum_{i=1}^n y_i$



68

Example: $\bar{x}$ is least squares

Given $x_1, x_2, \dots, x_n$, what value a minimizes the sum of squared differences between the x_i and a ?

$$\begin{aligned} & \min_a \sum_{i=1}^n (x_i - a)^2 \\ & \min_a \sum_{i=1}^n (x_i^2 - 2ax_i + a^2) \\ & \min_a \left(\sum_{i=1}^n x_i^2 - 2a \sum_{i=1}^n x_i + na^2 \right) \end{aligned}$$

69



Example: $\bar{x}$ is least squares

Take the derivative with respect to a and solve for a :

First order condition:

$$\begin{aligned} -2 \sum_{i=1}^n x_i + 2na &= 0 \\ 2na &= 2 \sum_{i=1}^n x_i \\ a &= \frac{1}{n} \sum_{i=1}^n x_i = \bar{x} \end{aligned}$$

70



Note the following, which are **not** properties of the summation operator:

$$\sum_{i=1}^n \frac{x_i}{y_i} \neq \frac{\sum_{i=1}^n x_i}{\sum_{i=1}^n y_i}$$

$$\sum_{i=1}^n x_i^2 \neq \left(\sum_{i=1}^n x_i \right)^2$$



71

Example of a double summation

Consider two sets of numbers $x_1, \dots, x_n$ and $y_1, \dots, y_n$. Use the index of summation i for x and the index j for y . The following is an example of a double summation:

$$\sum_{i=1}^n \sum_{j=1}^n x_i y_j$$

This can be written:

$$\sum_{i=1}^n x_i \sum_{j=1}^n y_j = x_1(y_1 + \dots + y_n) + x_2(y_1 + \dots + y_n) + \dots$$

Or:

$$x_1 y_1 + x_1 y_2 + x_1 y_3 + \dots + x_2 y_1 + x_2 y_2 + x_2 y_3 + \dots$$



72

Measures of variability



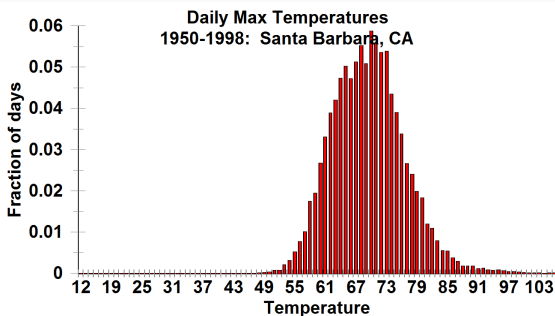
Measures of variability

Measures of **variability** are intended to capture how “spread out” the data points of the distribution are.

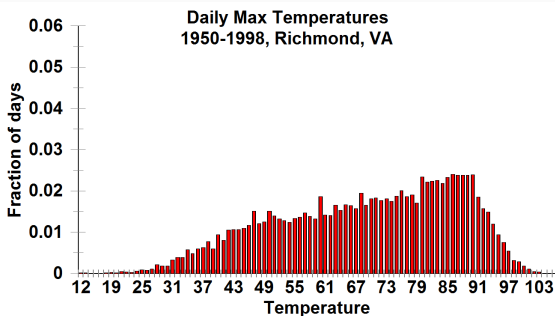
Two distributions can have the same measure of central tendency but very different dispersion. Example: consider mean daily high temperatures for two cities over 49 years:

- Richmond, VA: 69.0
- Santa Barbara: 69.5





74



75

Common measures of variability, dispersion, or “spread” include:

- Range
- Variance
- Standard deviation
- Interquartile range (IQR)



76

Range

The **range** is the difference between the largest and smallest observed values in a distribution

```
. sum achmat08, detail
```

math achievement in eighth grade				
	Percentiles	Smallest		
1%	38.55	36.61		
5%	41.89	37.14		
10%	44.185	37.2	obs	500
25%	49.42	37.24	Sum of wgt.	500
50%	56.18		Mean	56.59102
		Largest	Std. Dev.	9.339608
75%	63.74	77.2		
90%	68.935	77.2	variance	87.22827
95%	73.33	77.2	skewness	.1133238
99%	77.2	77.2	kurtosis	2.242742

For achmat08: $77.2 - 36.61 = 40.59$

77



The range is clearly sensitive to extreme values, since it is defined by the largest and smallest values.

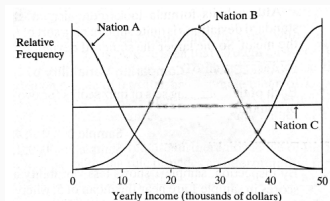
. sum expinc, detail					
expected income at age 30					
	Percentiles	Smallest			
1%	1	0			
5%	20000	0			
10%	25000	0	obs	459	
25%	30000	0	Sum of wgt.	459	
50%	40000		Mean	51574.73	
75%	55000	Largest	Std. Dev.	58265.76	
90%	80000	250000			
95%	100000	250000	Variance	3.39e+09	
99%	250000	500000	Skewness	10.89864	
		1000000	Kurtosis	162.8027	



78

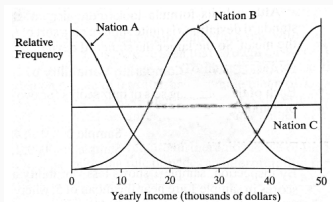
Range

Another disadvantage of the range is that it ignores all of the data beyond the maximum and minimum. In the following figure, the three income distributions have the same range (and mean):



79

- Nation B: incomes tend to be close to the mean (low variability)
- Nation A: incomes tend to be far from the mean (high variability)



80



Variance

A more meaningful measure of variability would capture the distance between *every* data point and the mean. Consider the “average deviation from the mean”:

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})$$

81



Table 5: Deviations above and below the mean

	GNI per cap	Math (x)	$x - \bar{x}$	Totals
Singapore	81,760	575	89.7	
Switzerland	110,330	508	22.7	
Canada	56,420	497	11.7	
Netherlands	68,530	493	7.7	
Ireland	87,360	492	6.7	
Denmark	77,190	489	3.7	
Belgium	59,500	489	3.7	
Australia	64,120	487	1.7	
Austria	60,360	487	1.7	149.0
Sweden	63,010	482	-3.3	
Germany	60,200	475	-10.3	
Norway	97,310	468	-17.3	
United States	88,810	465	-20.3	
Iceland	89,220	459	-26.3	
Qatar	74,330	414	-71.3	-149.0



82

Variance

- Problem: the sum of all deviations from the mean *will always be zero*—the negative deviations from the mean will cancel out the positive deviations.

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) = \frac{1}{n} \sum x_i - \frac{1}{n} n\bar{x} = \bar{x} - \bar{x} = 0$$

- Recall characterization of the mean as a “center of gravity,” where deviations above the mean are exactly balanced by deviations below the mean.



83

An alternative: the “average squared deviation from the mean,” or variance (s^2):

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

- the numerator is the **sum of squares**
- the denominator is $n - 1$ rather than 1 (explained later)



84

Variance

Table 6: Variance and standard deviation of PISA math scores

	GNI per cap	Math (x)	$x - \bar{x}$	$(x - \bar{x})^2$
Singapore	81,760	575	89.7	8040.1
Switzerland	110,330	508	22.7	513.8
Canada	56,420	497	11.7	136.1
Netherlands	68,530	493	7.7	58.8
Ireland	87,360	492	6.7	44.4
Denmark	77,190	489	3.7	13.4
Belgium	59,500	489	3.7	13.4
Australia	64,120	487	1.7	2.8
Austria	60,360	487	1.7	2.8
Sweden	63,010	482	-3.3	11.1
Germany	60,200	475	-10.3	106.8
Norway	97,310	468	-17.3	300.4
United States	88,810	465	-20.3	413.4
Iceland	89,220	459	-26.3	693.4
Qatar	74,330	414	-71.3	5088.4
ΣX		7280	0	15,439.3
Variance (s^2)				(15439/14)=1102.8
Std. Dev (s)				$\sqrt{1102.8} = 33.2$



85

- A variance of zero means there is no variation in x
- It is hard to interpret the variance in isolation, but it could be used to compare two distributions of a similar measure
- The magnitude of s^2 depends on the units of measurement. E.g.: variance of income in *cents* will be higher than the variance of income in *dollars*
- s^2 is generally hard to interpret (it is the average “squared deviation from the mean”)



86

Variance: why $n - 1$?

Why do we divide by $n - 1$ rather than n ?

- Typically we are *estimating* a population variance from a sample
- The variance formula includes $\bar{x}$ which is *also estimated* and may vary from sample to sample, in ways related to the variance.
- Dividing by n systematically *under-estimates* the variance by a factor of $(n - 1)/n$. It is *biased*.
- $n - 1$ is sometimes called the *degrees of freedom* since $\bar{x}$ provides one piece of information about the sample.
- If you are calculating variance for a *population*, dividing by n is appropriate.
- It is common practice just to divide by $n - 1$ in all cases. In large samples, it makes little difference.

For more see: <https://towardsdatascience.com/why-sample-variance-is-divided-by-n-1-89821b83ef6d>



87

The **standard deviation** (s) restores the variance measure to units of the original variable:

$$s = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

- If it helps, think of this as the “average” or “typical” deviation of x from its mean (though this is not strictly correct).
- s^2 and s are always positive, unless there is no variance in x (in which case they are both zero)



88

Standard deviation

Table 7: Variance and standard deviation of PISA math scores

	GNI per cap	Math (x)	$x - \bar{x}$	$(x - \bar{x})^2$
Singapore	81,760	575	89.7	8040.1
Switzerland	110,330	508	22.7	513.8
Canada	56,420	497	11.7	136.1
Netherlands	68,530	493	7.7	58.8
Ireland	87,360	492	6.7	44.4
Denmark	77,190	489	3.7	13.4
Belgium	59,500	489	3.7	13.4
Australia	64,120	487	1.7	2.8
Austria	60,360	487	1.7	2.8
Sweden	63,010	482	-3.3	11.1
Germany	60,200	475	-10.3	106.8
Norway	97,310	468	-17.3	300.4
United States	88,810	465	-20.3	413.4
Iceland	89,220	459	-26.3	693.4
Qatar	74,330	414	-71.3	5088.4
ΣX		7280	0	15,439.3
Variance (s^2)				(15439/14)=1102.8
Std. Dev (s)				$\sqrt{1102.8} = 33.2$



89

Standard deviation

The variance and standard deviation are obtained using `summarize` with the `detail` option:

```
. sum achmat08, detail
```

math achievement in eighth grade				
	Percentiles	Smallest		
1%	38.55	36.61		
5%	41.89	37.14		
10%	44.185	37.2	obs	500
25%	49.42	37.24	Sum of wgt.	500
50%	56.18		Mean	56.59102
		Largest	Std. Dev.	9.339608
75%	63.74	77.2		
90%	68.935	77.2	Variance	87.22827
95%	73.33	77.2	Skewness	.1133238
99%	77.2	77.2	Kurtosis	2.242742

Can use `summarize` alone for standard deviation. Also, `tabstat` with `stat(sd)`



90

Standard deviation

```
. sum expinc, detail
```

expected income at age 30				
	Percentiles	Smallest		
1%	1	0		
5%	20000	0		
10%	25000	0	obs	459
25%	30000	0	Sum of wgt.	459
50%	40000		Mean	51574.73
		Largest	Std. Dev.	58265.76
75%	55000	250000		
90%	80000	250000	Variance	3.39e+09
95%	100000	500000	Skewness	10.89864
99%	250000	1000000	Kurtosis	162.8027



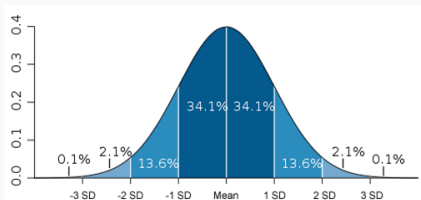
91

- The variance and standard deviation are especially sensitive to outliers/extreme values, since the variance formula *squares* deviations from the mean
- With **normal** (bell-shaped) distributions, s provides a good rule of thumb (the “Empirical Rule”):
 - About 68% of observations lie within 1 s.d. of the mean
 - About 95% of observations lie within 2 s.d. of the mean
 - Nearly all (99%) lie within 3 s.d. of the mean
 - Example: IQ mean of 100, standard deviation of 15



92

Standard deviation



93

The **coefficient of variation** (CV) expresses the standard deviation as a percentage of the mean:

$$CV = \frac{S}{\bar{X}} * 100$$

- Must have $\bar{x} \neq 0$. If $\bar{x} < 0$, can use the absolute value of $\bar{x}$.
- The CV adjusts for the scale of units used, allowing for more appropriate comparisons. For example, the CV is often used to measure inequality in school spending across states or districts. Comparing s across these groups would be misleading if the *level* of spending was higher in some districts/states than others.



94

Choosing a measure of variation

- Choice of dispersion measure should be made carefully—be aware of extreme values / outliers that may influence or distort the picture
 - The IQR is a “robust” alternative (shown later)
- One possibility when outliers exist: calculate statistics with outliers *dropped* (but be forthcoming about this—e.g. discuss both sets of results—and understand why outliers exist)



95

You may be asking why we square deviations from the mean (variance) and then take the square root (standard deviation) when we could work with absolute values:

$$MAD = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$$

You can do this, or (instead) calculate mean deviations from the *median*. (In Stata, the **egen** functions **mdev** and **mad** provide the mean deviation from the mean and median, respectively.)

These are rarely used, in part because the standard deviation has nice mathematical and statistical properties. For example, the sd is a defining feature of the normal distribution, while the MAD is not. But there is a “robust” debate over the value of the MAD.



Skewness



The **skewness statistic** is a summary of the positive or negative skewness of a distribution:

$$G = \frac{n}{(n-1)(n-2)} \frac{\sum_{i=1}^n (x_i - \bar{x})^3}{s^3}$$

- Notice this is based on *cubed* deviations from the mean (can be positive or negative).
- Positive for positively skewed distributions and negative for negatively skewed distributions (zero for symmetric)



97

Skewness statistic

How should we interpret this statistic? How do we know if a distribution is “significantly skewed”? As a rule of thumb, we can compare G to the **standard error of the skewness**:

$$\sqrt{\frac{6n(n-1)}{(n-2)(n+1)(n+3)}}$$

With large enough samples, this is approximately:

$$\sqrt{6/n}$$



98

For now: if the skewness statistic is more than 2x the standard error, we can say the distribution is “significantly skewed.”

- The concept of a standard error is introduced later
- Can calculate yourself in Stata using `display`, or install the user-written `summskew` command (see Github for ado file).

Aside: this is essentially a test against a *normal* symmetric distribution. If your distribution departs significantly from normality, this may not be the best approximation. See Wright & Herrington (2011).



99

Skewness statistic in Stata

The skewness statistic is reported using `summarize` with `detail`:

```
. sum achmat08, detail
```

math achievement in eighth grade					
	Percentiles	Smallest			
1%	38.55	36.61			
5%	41.89	37.14			
10%	44.185	37.2	Obs		500
25%	49.42	37.24	Sum of wgt.		500
50%	56.18		Mean	56.59102	
75%	63.74	Largest	Std. Dev.	9.339608	
90%	68.935	77.2			
95%	73.33	77.2	Variance	87.22827	
99%	77.2	77.2	Skewness	.1133238	
			Kurtosis	2.242742	

- `achmat08` has a slight positive skew
- Std err of skewness not reported here by Stata, but `display sqrt((6*500*499)/(498*501*503))` yields 0.109. The skewness statistic is *not* more than twice the std err of the skewness



Quantiles and measures of relative position



Quantiles

Quantiles are cutpoints that divide a distribution into continuous intervals of equal size. Common quantiles include:

- **Quartiles:** 4 equally sized groups
- **Deciles:** 10 equally sized groups
- **Terciles:** 3 equally sized groups
- **Percentiles:** 100 equally sized groups

If there are n observations in a dataset, *quantiles* will include $n/4$ observations in each group. Note the term “quantile” is often used to refer to both the cutpoint and the groups themselves.



In real datasets, there may not be the right number of data points to divide into equally sized quantiles. For example, if $n = 10$, you can't divide the 10 observations into 4 equally sized groups. So what rule do you use?

If dividing the data into q groups, and you want to find the k th quantile ($k \leq q$), find the observation ranked $n * k/q$ when ranking the data points from smallest to largest.

- If this is an *fractional* value, take the next-largest ranked observation.
- If this is an *integer* value, take the midpoint of that ranked observation and the next-largest one.



102

Example: suppose $n = 74$

- 5th percentile is the $74 * (5/100) = 3.7$ th obs (round to 4)
- 10th percentile is the $74 * (10/100) = 7.4$ th obs (round to 8)
- 25th percentile is the $74 * (25/100) = 18.5$ th obs (round to 19)
- Median is the $74 * (50/100) = 37$ th obs (take avg of 37th and 38th)

This operationalizes the p th percentile as the smallest value that is greater than $p\%$ of the observations.



103

Can sometimes determine percentiles from cumulative frequency distributions:

```
. tabulate fansize
```

family size	Freq.	Percent	Cum.
2	9	1.80	1.80
3	52	10.40	12.20
4	199	39.80	52.00
5	142	28.40	80.40
6	55	11.00	91.40
7	21	4.20	95.60
8	9	1.80	97.40
9	13	2.60	100.00
Total	500	100.00	

e.g., 10th percentile is $500 \times (10/100) = 50$ th obs (take avg of 50th and 51st) = 3



104

Percentiles

For quantitative variables, can use **summarize** with **detail** option in Stata to get common percentiles:

```
. summ achrdd08, detail
```

reading achievement in eighth grade					
Percentiles		Smallest			
1%	37.525	35.74			
5%	40.789	35.82			
10%	43.71	37.17	Obs		500
25%	49.975	37.29	Sum of wgt.		500
50%	56.445		Mean	56.04986	
		Largest	Std. dev.	8.829726	
75%	63.195	70.55			
90%	69.15	70.55	Variance	77.96406	
95%	70.55	70.55	Skewness	-.1485071	
99%	70.55	70.55	Kurtosis	2.191284	

Alternative 1: **tabstat** with **stat(p25 p50 etc)**.

Alternative 2: create a variable that contains a percentile of *varname* using **egen**. E.g., **egen varnamep10=pctile(varname), p(10)**



105

A third alternative uses `centile varname, centile(p)`.

Note, however, that `centile` uses a linear interpolation rather than the “next highest rank” method. This will often provide a slightly different answer.

- `centile` will calculate the rank $P = (n + 1)p/100$ where p is the desired percentile.
- If P is fractional, `centile` will interpolate between the two observations ranked $\text{int}(P)$ and $\text{int}(P) + 1$.
- If P is an integer, it will use that ranked observation for the p th percentile.



106

If you would like to create a new variable in your dataset that contains quantile values, see the `pctile` command:

```
pctile achmat08d = achmat08, nq(10) genp(ptile)
list achmat08d ptile in 1/10
```

If you would like to create a new variable in your dataset that contains the quantile each observation falls within, see the `xtile` command:

```
xtile achmat08q = achmat08, nq(4)
tabulate achmat08q
```



107

The **five number summary** is a report of the minimum, Q1, median, Q3, and maximum. In Stata, use `summarize`, `detail` or `tabstat`:

```
. tabstat achrdg08 achmat08 achsci08 achsls08, stat(min p25 p50 p75 max)
```

stats	achrdg08	achmat08	achsci08	achsls08
min	35.74	36.61	34.94	29.2
p25	49.975	49.42	48.69	49.39
p50	56.445	56.18	55.4	54.76
p75	63.195	63.74	62.19	61.2
max	70.55	77.2	80.01	76.7

108



The p th percentile is sometimes defined as the value that is greater than or equal to $p\%$ of the observations (vs. “greater than”).

This can make a difference in some datasets. In the limit—with continuous distributions—there will be no difference in these definitions.

109



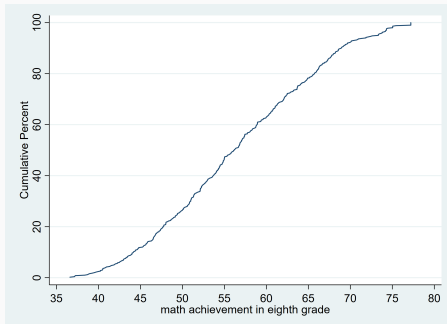
Empirical CDF (“ogive”)

- A graph showing percentiles on the vertical axis and values of x on the horizontal axis is an **ogive** (OH-jive), or **empirical CDF** (cumulative distribution function)
- This graph in Stata requires the extra step of creating a new variable with the cumulative relative frequency, using `cumul`:
 - `cumul achmat08, gen(cumachmat08)`
 - `replace cumachmat08=cumachmat08*100`
 - `sort cumachmat08`
 - `twoway line cumachmat08 achmat08, ytitle("`Cumulative Percent'')`
- Alternative: `ssc install distplot`



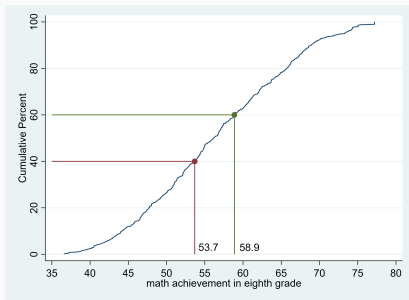
110

Empirical CDF (“ogive”)



111

Empirical CDF (“ogive”)



- The 40th percentile is approximately 53.7
- The 60th percentile is approximately 58.9

112



Z-scores

Percentiles are a measure of *position* in a distribution. Another is the **z-score** (or standardized score). The z-score for a particular value of x (x_i) is defined as:

$$z_i = \frac{x_i - \bar{X}}{s}$$

(Details later in this lecture)

113



- The **interquartile range** is the difference between the upper and lower quartiles (75th and 25th percentiles)—a measure of dispersion that is robust to extreme values/outliers
- Pull quartiles from `summarize`, `detail` or `tabstat`, or use the `centile` command.
- You can also use the `iqr` stat option in `tabstat`



114

Interquartile range

```
. tabstat achrdg08 achmat08 achsci08 achsls08, stat(min p25 p50 p75 max)
```

stats	achrdg08	achmat08	achsci08	achsls08
min	35.74	36.61	34.94	29.2
p25	49.975	49.42	48.69	49.39
p50	56.445	56.18	55.4	54.76
p75	63.195	63.74	62.19	61.2
max	70.55	77.2	80.01	76.7


$$\text{IQR} = 63.195 - 49.975 = 13.22$$


$$\text{IQR} = 61.2 - 49.39 = 11.81$$



115

Compare the IQR (\$25,000) to s (\$58,265)

```
. sum expinc30,detail
```

expected income at age 30					
Percentiles		Smallest			
1%	1	0			
5%	20000	0			
10%	25000	0	Obs		459
25%	30000	0	Sum of wgt.		459
50%	40000		Mean		51574.73
		Largest	Std. dev.		58265.76
75%	55000	250000			
90%	80000	250000	Variance		3.39e+09
95%	100000	500000	Skewness		10.89864
99%	250000	1000000	Kurtosis		162.8027

116



Box plots

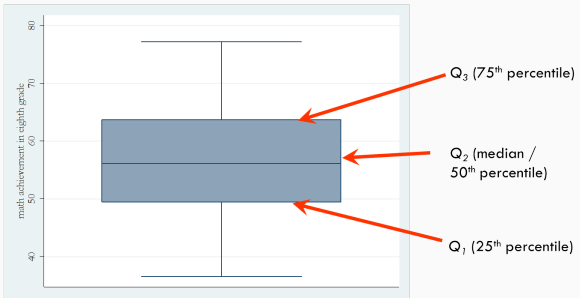
A **boxplot** (or box-and-whiskers plot) shows features of a variable's distribution: median, Q3, Q1, and extreme values (tails)

- The tails are “whiskers” that extend to the maximum and minimum in the data *if* there are no outliers (defined here as 1.5 IQR lengths above Q3 or below Q1).
- If there are outliers, the whiskers extend to the observation closest to (but not beyond) these thresholds (the highest and lowest **adjacent value**). Outliers are shown as points or asterisks (*).
- In Stata: `graph box varname`

117



Box plots



- Median=56.18, Q1=49.42, Q3=63.74, IQR=14.32
- Outliers would be above $(63.74 + (1.5 * 14.32)) = 85.22$ or below $(49.42 - (1.5 * 14.32)) = 27.94$



118

Box plots



- Outliers would be above $(55,000 + (1.5 * 25,000)) = \$92,500$ or below $(30,000 - (1.5 * 25,000)) = -\$7,500$



119

- Stata can suppress the outliers with the `nooutliers` option

Transforming variables



Transforming variables

Transforming a variable simply means expressing it another way. Useful in many circumstances:

- Changing units of measure (e.g. feet to inches)
- Putting an observed value into the context of its distribution (e.g., “is 160 a high LSAT score?”)
- Reduce the skewness of a variable—to work with a more symmetric variable
- Compare distributions on different scales
- Combining multiple variables into one index



Some examples from education research:

- Convert counts into a rate or percentage (e.g., attendance rate)
- Create a binary indicator from a continuous variable (e.g., chronically absent)
- Center a variable around its mean
- Log of enrollment
- Create a change score (e.g., test score growth)
- Nonlinear terms in a regression analysis (e.g., enrollment squared)



121

Transforming variables

The z-score was one example of a transformation: from the units of the original scale to “standard deviation units”

- Example: US PISA math score is 465 while the mean (richest 15 countries) is 485.3. The US z-score is $(465 - 485.3)/33.2 = -0.611$. US scores are 0.611 standard deviations below the mean.



122

- Transformations can be linear or nonlinear
- Monotonic transformations preserve the *order* of the data points
- Non-monotonic transformations fail to preserve the order of the data points. Example: x^2 , winsorizing
- We will only consider the monotonic transformations here.



123

Transforming variables

A **linear transformation** involves only addition, subtraction, multiplication, or division, applied to the original variable x_0 :

$$x_1 = a + (b * x_0)$$

- a can be positive (addition) or negative (subtraction)
- $b \neq 0$, but can be negative and $|b|$ can be < 1 (i.e., division)



124

In Stata, **generate** (or **gen**) is the all-purpose command for creating new variables. Example: expected income, in thousands of dollars

```
. gen expinc30b = (expinc30 / 1000)  
(41 missing values generated)
```

```
. sum expinc*
```

Variable	Obs	Mean	Std. Dev.	Min	Max
expinc30	459	51574.73	58265.76	0	1000000
expinc30b	459	51.57473	58.26576	0	1000

125



Translation

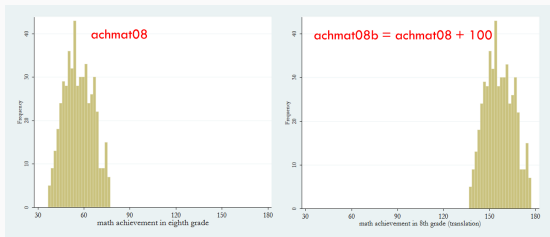
The simplest linear transformation is a **translation** where $b = 1$ and a is positive or negative:

$$X_1 = a + X_0$$

Shifts the distribution to the right or left. E.g. `achmat08r=achmat08 + 100`

126





127



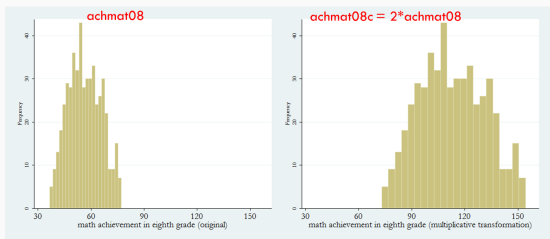
Multiplicative transformation

- The translation did not affect the *shape* or *spread* (variability) of the distribution, but did affect its position, shifting it to the left or right.
- Now consider a **multiplicative** transformation, where $a = 0$ but $b \neq 0$ and $b \neq 1$:

$$x_1 = (b * x_0)$$

128





129



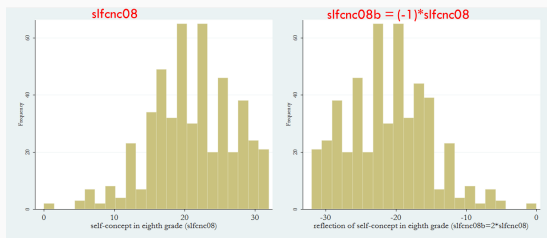
Multiplicative transformation

- The multiplicative transformation has affected *both* the location of the distribution *and* its spread.
 - The distribution is “stretched” whenever $|b| > 1$
 - The distribution is “compressed” whenever $|b| < 1$
- While the spread is affected, *relative* distance between points is preserved. *Shape* (symmetry, skewness) is preserved unless $b < 0$, in which case the distribution is flipped to its mirror image. A **reflection** multiplies the original variable by -1 .

130



Reflection:



131



Transforming variables

How will a linear transformation of a variable affect its:

- Mean?
- Variance?
- Standard deviation?
- Range?
- IQR?
- Skewness?

132



How a linear transformation affects the mean ($\bar{x}_{new}$ vs. $\bar{x}_{old}$):

$$\bar{x}_{old} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\bar{x}_{new} = \frac{1}{n} \sum_{i=1}^n (a + bx_i)$$

$$\bar{x}_{new} = \frac{1}{n} \sum_{i=1}^n a + \frac{1}{n} \sum_{i=1}^n bx_i$$

$$\bar{x}_{new} = a + b\bar{x}_{old}$$



133

In other words, the mean of the new variable ($\bar{x}_{new}$) is equal to the old mean ($\bar{x}_{old}$) multiplied by b , and increased (or decreased) by a . Also true for median, mode, and percentiles.



134

Example: use a linear transformation to express Fahrenheit temperature in Celsius.

- The conversion from F to C is: $x_C = -17.78 + 0.556 * x_F$ (a linear transformation)
- The mean daily high in Santa Barbara (in Fahrenheit) is $\bar{x}_F = 69.5$
- So then: $\bar{x}_C = -17.78 + 0.556 * 69.5 = 20.86$



135

Transforming variables

How a linear transformation affects the variance (s_{new}^2 vs. s_{old}^2):

$$s_{old}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s_{new}^2 = \frac{1}{n-1} \sum_{i=1}^n (a + bx_i - (a + b\bar{x}_{old}))^2$$

$$s_{new}^2 = \frac{1}{n-1} \sum_{i=1}^n (bx_i - b\bar{x}_{old})^2$$



136

Continued:

$$s_{new}^2 = \frac{1}{n-1} \sum_{i=1}^n b^2 (x_i - \bar{x}_{old})^2$$

$$s_{new}^2 = b^2 \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}_{old})^2$$

$$s_{new}^2 = b^2 s_{old}^2$$



137

Transforming variables

- In other words, the variance of the transformed variable (s_{new}^2) is equal to the variance of the original variable (s_{old}^2), multiplied by b^2 .
- The standard deviation of the transformed variable is:

$$s_{new} = \sqrt{s_{new}^2} = \sqrt{b^2 s_{old}^2} = |b| s_{old}$$

- Notice a does not affect the variance or standard deviation.
- The range and IQR of the transformed variable are equal to the range and IQR of the original variable, multiplied by $|b|$



138

The skewness statistic is unaffected by a linear transformation *unless* b is negative. Then the skewness of the transformed variable is $(-1) * G_{old}$



139

Standardized variables

Standardized variables are an example of a linear transformation

In general a standardized variable applies a linear transformation to a variable to give it a predefined mean (c) and standard deviation ($v > 0$)

	Variable	Mean
Original	x	$\bar{x}$
Transformed	$x - \bar{x} + c$	c

	Variable	Standard deviation
Original	x	s
Transformed	$\frac{v}{s}x$	v



140

Examples of standardized variables:

- SAT sections: mean 500, sd 100
- GRE sections: 130-170, mean 150, sd 8
- IQ score: mean 100, sd 15
- MCAT section: mean 125, sd 3
- z-score: mean 0, sd 1



141

Z-score transformation

It is easy to go from standard score to original scale score, given the mean and sd:

$$Z = \frac{X - \bar{X}}{S}$$

$$X = \bar{X} + SZ$$

Example:

- *achtmat08*: mean 56.59 and sd 9.34
- $z = 0.89$ would be $x = 56.59 + 9.34(0.89) = 64.9$



142

There are several ways to obtain z-scores in Stata. Manually, can obtain mean and sd, and then create the z-score using `gen`

```
. sum expinc30
```

Variable	Obs	Mean	Std. dev.	Min	Max
expinc30	459	51574.73	58265.76	0	1000000

```
. sum achmat08
```

Variable	Obs	Mean	Std. dev.	Min	Max
achmat08	500	56.59102	9.339608	36.61	77.2

```
. gen zachmat08 = (achmat08 - 56.59102) / 9.339608
```

```
. sum zachmat08
```

Variable	Obs	Mean	Std. dev.	Min	Max
zachmat08	500	1.04e-08	1	-2.139385	2.206621

Can also use `egen` with `std` function (*preferable*)



143

Z-score transformation

Alternatively, `sum` saves its results in “`r()`” macros:

```
. sum achmat08
```

Variable	Obs	Mean	Std. dev.	Min	Max
achmat08	500	56.59102	9.339608	36.61	77.2

```
. return list
```

Scalars:

```

r(N) = 500
r(sum_w) = 500
r(mean) = 56.59102010345459
r(Var) = 87.22826930507507
r(sd) = 9.339607556266756
r(min) = 36.61000061035156
r(max) = 77.19999694824219
r(sum) = 28295.51005172729

```

```
. gen zachmat08b = (achmat08 - r(mean)) / r(sd)
```

```
. summ zachmat08b
```

Variable	Obs	Mean	Std. dev.	Min	Max
zachmat08b	500	-9.77e-10	1	-2.139385	2.206621



144

- A *nonlinear* transformation involves some mathematical operation other than addition, subtraction, multiplication, and division
- There are lots of examples, but common ones include logarithmic (log), square root, inverse hyperbolic sine transformations



145

Nonlinear transformations

- Nonlinear transformations alter the shape (symmetry, skewness) of a distribution. In fact, this is often one of the purposes of a nonlinear transformation: to reduce the skewness in a distribution
- Many statistical procedures assume the variable has a normal or approximately normal distribution. In practice this assumption is violated. Transformations can help.



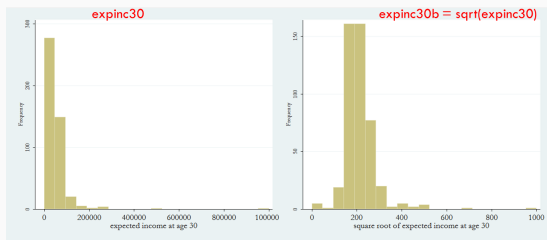
146

- Consider the square root transformation of expected income at age 30: `gen expinc30b=sqrt(expinc30)`
- Skewness statistics: $G_{old} = 10.898$, $G_{new} = 3.7$



147

Square root transformation



148

- The square root transformation reduces the magnitude of large values of the original variable by more than it does small values. Examples:
- Original value of 4 becomes 2 (half the size)
- Original value of 100 becomes 10 (1/10 the size)



- The square root is not defined for negative numbers. If the original variable has negative values, one can first apply a translation ($+a$) that ensures the minimum value is above zero (preferably equal to 1)
- Values between 0 and 1 become larger when applying the square root, unlike values above 1
- Log transformations have a similar effect on the distribution of a variable as the square root



- Recall that logarithms are exponents and depend on the base
- e.g. $\log_2 8 = 3$
- In this example, 2 is the base. $\log_2 8$ asks: “to what power does one take 2 to get 8?”
- The base can be any positive number other than 1



151

Log transformation

x value	Base 2	“log points”	x value	Base 10	“log points”
2	$\log_2 2$	1	10	$\log_{10} 10$	1
4	$\log_2 4$	2	100	$\log_{10} 100$	2
8	$\log_2 8$	3	1000	$\log_{10} 1000$	3
16	$\log_2 16$	4	10000	$\log_{10} 10000$	4
32	$\log_2 32$	5			
64	$\log_2 64$	6			



152

- On the $\log_2$ scale: each point is associated with 2 to that power
 - Values of 2, 4, 8, 16, ... $\rightarrow$ 1, 2, 3, 4
 - Larger values reduced more than smaller ones
- On the $\log_{10}$ scale: each point is associated with 10 to that power
 - Values of 10, 100, 1000, ... $\rightarrow$ 1, 2, 3
 - Transformation has a larger effect



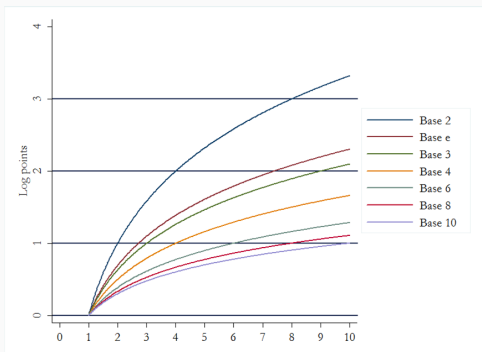
153

Log transformation

- Notice it takes larger and larger increases in x to go up by one unit (one “log point”). The square root has a similar property.
- The “natural logarithm” is a commonly used log function, where the base is 2.718 (e). The natural log is denoted $\ln(x)$.
- $\ln$ is so commonly used that whenever you hear that a “log” has been applied, it is almost always $\ln$.
- The natural log has a nice interpretation: a 0.01 change is approximately 1% change in the original variable. (This works well for small changes, but not larger ones).
 - More precisely: an increase of p in the log is equal to a $(e^p - 1) \times 100\%$ increase in the original variable. For small p , $(e^p - 1)$ and p are very similar.



154



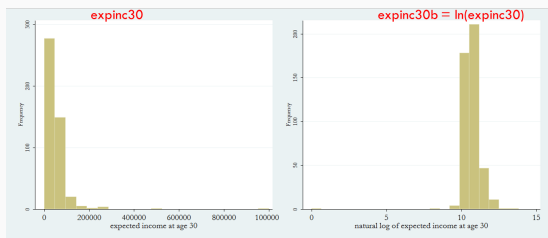
155



- Note the logarithm is only valid for $x \geq 1$. (The square root was only valid for $x \geq 0$)
- If x doesn't meet this condition, you can do a simple linear transformation first (e.g. add a constant a to all values such that the minimum $x \geq 1$).
- Another option: inverse hyperbolic sine transformation

156





Skewness statistics: $G_{old} = 10.898$, $G_{new} = -6.76$



157

Inverse hyperbolic sine transformation

The inverse hyperbolic sine function (or *arcsinh*) is defined both for $x > 0$ and $x \leq 0$:

$$IHS = \ln(x + \sqrt{x^2 + 1})$$

x	ln x	asinh x
-10	undef	-3.00
-5	undef	-2.31
0	undef	0.00
5	1.61	2.31
10	2.30	3.00
100	4.61	5.30
1000	6.91	7.60
50000	10.82	11.52
100000	11.51	12.21

In Stata, the IHS function is `asinh()`. See Bellemare & Wichman (2020) for more on IHS.



Other types of transformations:

- “collapsing” or re-coding variables—creating a categorical variable from a continuous variable with many possible values
- ranking
- combining variables—e.g. a composite score (math + reading + social studies + science) or an index of multiple outcomes (Kling, Liebman, & Katz, 2007)

Note: be aware of how missing values affect your calculation!



159

Recoding into ordinal groups

Example: *unitmath*, units of math classes taken, ranges from 1-6 but has fractional values. Can set up groups/intervals

```
. table unitmath
```

	Frequency
units in mathematics (naep)	
1	1
1.5	2
1.75	1
2	36
2.38	1
2.5	10
2.75	1
2.81	1
2.97	1
3	121
3.5	29
3.6	1
3.63	1
3.96	1
4	229
4.08	1
4.29	1
4.5	22
4.99	2
5	31
5.5	2
5.99	1
6	4
Total	500



160

```
. recode unitmath (1/1.99 = 1) (2/2.99 = 2) (3/3.99 = 3) (4/4.99 =  
4) (5/100 = 5), gen(unitmathc)  
(82 differences between unitmath and unitmathc)
```

```
. table unitmathc
```

```
-----  
RECODE of |  
unitmath |  
(units in |  
mathemati |  
cs |  
(naep)) | Freq.  
-----+-----  
1 | 4  
2 | 50  
3 | 153  
4 | 255  
5 | 38  
-----
```



161

Multiple outcome index

Kling, Liebman, & Katz (2007) looked at the effects of the Moving to Opportunity (MTO) experiment in which families in public housing were given Section 8 vouchers to relocate to lower-poverty neighborhoods. There are lots of possible outcomes which they combine them into indices:

- Index of economic self-sufficiency: equally weighted average of z-score for five measures of earnings, employment, and public assistance receipt
- Index of physical health (separate for adults and children): e.g., distress, depression, worrying, calmness, sleep
- Index of mental health (separate for adults and children): e.g., self-reported health, asthma, obesity, hypertension
- Index of youth risky behavior: e.g., marijuana, smoking, alcohol
- Index of youth education: e.g., graduated, reading score, math score



162

Winsorizing is a type of non-monotonic transformation—it is a way of addressing outliers.

- **Trimming** means discarding (or ignoring) outlier observations, for example the bottom and top 1% of values.
- **Winsorizing** means re-coding outlier observations to less extreme values (e.g., setting values above the 95th percentile to the 95th percentile)

See Stata packages `trimmean`, `trimplot` and `winsor`, but use with good judgment! Removing/re-coding outliers has consequences.



Next time



- Basic rules of probability; probability as relative frequency
- Conditional probability and Bayes' Rule
- Probability distributions: discrete and continuous
- Binomial distribution
- Normal distribution
- And much, much, more!

